GATE CS — Databases (DBMS)

Topic-wise past questions (2016–2026) + new practice questions with solutions

Built from the papers in pAPER/ of the GATE2027 repo. Topics are ordered by total marks (most important first).

Topic	Past questions	Total marks	New practice Qs
SQL, relational algebra & calculus	22	35	2
Normalization & FDs	17	27	2
Transactions & concurrency control	15	23	2
Indexing, B/B+ trees, file organization	9	14	2
ER model	6	6	2
Total	**69**	**105**	**10**

How to read this

- Each topic has **A. Past-paper questions** (newest first; tagged year · set · question number · marks) and **B. New practice questions** with short solutions.
- Past questions have **no answer key** here; check them against the official answer keys.
- Question text was extracted from the PDFs. Code, tables, symbols and figures may be flattened or lost. Entries marked [OCR] come from scanned papers and may contain errors; open the original in pAPER/<year>/ for the exact wording.
- New practice questions are original, in GATE style; numerical and program-output answers were verified.

SQL, relational algebra & calculus

SQL / relational algebra / tuple calculus query results. · 22 questions, 35 marks over 15 papers (≈2.3 marks/paper)

A. Past-paper questions (22)

2026 CS1 · Q.43 · 2 marks
Consider a relational database schema with two relations R(P, Q) and S(X, Y). Let E = {⟨u⟩ | ∃v ∃w ⟨u, v⟩ ∈ R ∧ ⟨v, w⟩ ∈ S} be a tuple relational calculus expression. Which one of the following relational algebraic expressions is equivalent to E ?
(A) ΠP (R ⋈R.P=S.X S)
(B) ΠP (S ⋈S.X=R.Q R)
(C) ΠP (R ⋈R.P=S.Y S)
(D) ΠP (S ⋈S.Y=R.Q R)

2025 CS1 · Q.39 · 2 marks
Consider two relations describing teams and players in a sports league: • teams(tid, tname): tid, tname are team-id and team-name, respectively • players(pid, pname, tid): pid, pname, and tid denote player-id, player- name and the team-id of the player, respectively Which ONE of the following tuple relational calculus queries returns the name of the players who play for the team having tname as ′MI′?
(A) { p. pname | p ∈ players ∧ ∃t (t ∈ teams ∧ p. tid = t. tid ∧ t. tname = ′MI′)}
(B) { p. pname | p ∈ teams ∧ ∃t (t ∈ players ∧ p. tid = t. tid ∧ t. tname = ′MI′)}
(C) { p. pname | p ∈ players ∧ ∃t (t ∈ teams ∧ t. tname = ′MI′)}
(D) { p. pname | p ∈ teams ∧ ∃t (t ∈ players ∧ t. tname = ′MI′)}

2025 CS1 · Q.55 · 2 marks
Consider the following database tables of a sports league. player(pid,pname,age) team(tid,tname,city,cid) coach(cid,cname) members(pid,tid) An instance of the table and an SQL query are given. player coach team members SELECT MIN(P.age) FROM player P WHERE P.pid IN (SELECT M.pid FROM team T, coach C, members M WHERE C.cname = 'Mark' AND T.cid = C.cid AND M.tid = T.tid) The value returned by the given SQL query is _____ . (Answer in integer)

2025 CS2 · Q.54 · 2 marks
Consider the following relational schema: Students (rollno: integer, name: string, age: integer, cgpa: real) Courses (courseno: integer, cname: string, credits: integer) Enrolled (rollno: integer, courseno: integer, grade: string) Which of the following options is/are correct SQL query/queries to retrieve the names of the students enrolled in course number (i.e., courseno) 1470?
(A) SELECT S.name FROM Students S WHERE EXISTS (SELECT * FROM Enrolled E WHERE E.courseno = 1470 AND E.rollno = S.rollno);
(B) SELECT S.name FROM Students S WHERE SIZEOF (SELECT * FROM Enrolled E WHERE E.courseno = 1470 AND E.rollno = S.rollno) > 0;
(C) SELECT S.name FROM Students S WHERE 0 < (SELECT COUNT(*) FROM Enrolled E WHERE E.courseno = 1470 AND E.rollno = S.rollno);
(D) SELECT S.name FROM Students S NATURAL JOIN Enrolled E WHERE E.courseno = 1470;

2024 CS1 · Q.35 · 1 mark
Consider the following two relations, R(A,B) and S(A,C): R S A B A C 10 20 10 90 20 30 30 45 30 40 40 80 30 50 50 95 The total number of tuples obtained by evaluating the following expression σB<C (R ⋈R.A=S.A S) is _____

2024 CS2 · Q.45 · 2 marks
The relation schema, Person(pid,city), describes the city of residence for every person uniquely identified by pid. The following relational algebra operators are available: selection, projection, cross product, and rename. To find the list of cities where at least 3 persons reside, using the above operators, the minimum number of cross product operations that must be used is
(A) 1
(B) 2
(C) 3
(D) 4

2023 · Q.16 · 1 mark
Which one of the options given below refers to the degree (or arity) of a relation in relational database systems?
(A) Number of attributes of its relation schema.
(B) Number of tuples stored in the relation.
(C) Number of entries in the relation.
(D) Number of distinct domains of its relation schema. CS

2023 · Q.61 · 2 marks
Consider the following table named Student in a relational database. The primary key of this table is rollNum. Student rollNum name gender marks 1 Naman M 62 2 Aliya F 70 3 Aliya F 80 4 James M 82 5 Swati F 65 The SQL query below is executed on this database. SELECT * FROM Student WHERE gender = ‘F’ AND marks > 65; The number of rows returned by the query is . CS

2022 · Q.14 · 1 mark
In a relational data model, which one of the following statements is TRUE?
(A) A relation with only two attributes is always in BCNF.
(B) If all attributes of a relation are prime attributes, then the relation is in BCNF.
(C) Every relation has at least one non-prime attribute.
(D) BCNF decompositions preserve functional dependencies.

2022 · Q.25 · 1 mark
Consider the following three relations in a relational database. Employee(eId , Name), Brand (bId , bName), Own(eId , bId) Which of the following relational algebra expressions return the set of eIds who own all the brands?
(A) eId (eId ,bId (Own) / bId (Brand))
(B) eId (Own) eId (eId (Own) bId (Brand)) eId ,bId (Own)
(C) eId (eId ,bId (Own) / bId (Own))
(D) eId (eId (Own) bId (Own)) / bId (Brand)

2022 · Q.56 · 2 marks
Consider the relational database with the following four schemas and their respective instances. Student(sNo, sName, dNo) Dept(dNo, dName) Course(cNo, cName, dNo) Register(sNo, cNo) SQL Query: SELECT * FROM Student AS S WHERE NOT EXIST (SELECT cNo FROM Course WHERE dNo = “D01” EXCEPT SELECT cNo FROM Register WHERE sNo = S.sNo) The number of rows returned by the above SQL query is_____.

2021 · Q.23 · 1 mark [OCR]
A relation r(A, B) in a relational database has 1200 tuples. The attribute A has integer values ranging from 6 to 20, and the attribute B has integer values ranging from 1 to 20. Assume that the attributes A and B are independently distributed. The estimated number of tuples in the output of 0(.4310)v(e=1s)(7) is CS, Set-1 - Copyright © GATE 2021

2020 · Q.13 · 1 mark [OCR]
Consider a relational database containing the following schemas. Catalogue Suppliers sno | pno cost sno | sname location Sl Pl 150 S1__| M/s Royal furniture Delhi Sl P2 50 S2__| M/s Balaji furniture Bangalore Sl P3 100 S3__| M/s Premium furniture Chennai S2 P4 200 Parts S2 PS 250 5 ea Pi 50 pno | pname part_spec $3 P2 150 P1_| Table Wood $3 PS 300 P2_| Chair Wood $3 P4 250 P3_| Table Steel P4_| Almirah Steel PS__| Almirah Wood The primary key of each table is indicated by underlining the constituent fields. SELECT s.sno, s.sname FROM Suppliers s, Catalogue c WHERE s.sno = c.sno AND cost > (SELECT AVG (cost) FROM Catalogue WHERE pno = ‘P4’ GROUP BY pno); The number of rows returned by the above SQL query is
(A) 4 (6) 5 (©) 0 (0) 2

2019 · Q.51 · 2 marks [OCR]
A relational database contains two tables Student and Performance as shown below: Student Performance Roll_no. | Student_name Roll_no. | Subject_code | Marks 1 Amit 1 A 86 2 Priya 1 B 95 3 Vinit 1 Cc 90 4 Rohan 2 A 89 5 Smita 2 Cc 92 3 Cc 80 The primary key of the Student table is Roll_no. For the Performance table, the columns Roll_no. and Subject_code together form the primary key. Consider the SQL query given below: SELECT S.Student_name, sum(P.Marks) FROM Student S, Performance P WHERE P.Marks > 84 GROUP BY S.Student_name; The number of rows returned by the above SQL query is

2019 · Q.55 · 2 marks [OCR]
Consider the following relations P(X, Y,Z), Q(X, Y,T) and R(Y,V). P Q R x Y Z x Y |T Y Vv X1 | Yl {| Zi X2 | YI | 2 Yl {| Vi x1 | yl | z2 X1 | Y2 | 5 Y3 | V2 x2 | v2 | 72 X1 | YI | 6 Y2 | V3 x2 | v4 | zZ4 X3 | Y3] 1 Y2 | V2 How many tuples will be returned by the following relational algebra query? I] (ow@y=ry ARV=Vv2) (Px R)) - I] (@r=ry ner>2) (Qx R)) Xx x Answer: cs 13/15 Q.No. Type Section Key Marks 1 McQ GA Cc 1 2 McQ GA A 1 3 McQ GA B 1 4 McQ GA Cc 1 5 McQ GA D 1 6 McQ GA B 2 7 MCQ GA Cc 2 8 McQ GA B 2 9 McQ GA Cc 2 10 McQ GA D 2 1 McQ cs D 1 2 MCQ cs A 1 3 MCQ cs D 1 4 McQ cs Cc 1 5 McQ cs Cc 1 6 MCQ cs B 1 7 MCQ cs B 1 8 McQ cs Cc 1 9 McQ cs D 1 10 McQ cs B 1 11 MCQ cs Cc 1 12 MCQ cs CORD 1 13 MCQ cs Cc 1 Q.No. Type Section Key Marks 14 McQ cs B 1 15 MCQ cs D i 16 MCQ cs B 1 17 NAT cs 31 to 31 1 18 NAT cs 26 to 26 1 19 NAT cs 31 to 31 1 20 NAT cs 0.08 to 0.08 1 21 NAT cs 2to2 1 22 NAT cs 0.502 to 0.504 1 23 NAT cs 80 to 80 1 24 NAT cs 6to6 1 25 NAT cs 29 to 29 1 26 MCQ cs D 2 27 MCQ cs B 2 28 MCQ cs Cc 2 29 MCQ cs Cc 2 30 MCQ cs A 2 31 MCQ cs Cc 2 32 MCQ cs Cc 2 33 McQ cs B 2 34 MCQ cs B 2 35 MCQ cs Cc 2 36 MCQ cs A 2 Q.No. Type Section Key Marks 37 McQ cs Cc 2 38 MCQ cs c 2 39 MCQ cs A 2 40 McQ cs A 2 41 NAT cs 2to2 2 42 NAT cs 3.7 to 3.8 OR 4.0to 4.1 2 43 NAT cs 5to5 2 44 NAT cs 12 to 12 2 45 NAT cs 160 to 160 2 46 NAT cs 4.25 to 4.25 2 47 NAT cs 0.8 to 0.8 2 48 NAT cs 120 to 120 2 49 NAT cs 3to3 2 50 NAT cs 3 to3 2 51 NAT cs 5to5 2 52 NAT cs 5to5 2 53 NAT cs 10 to 10 2 54 NAT cs 97 to 97 2 55 NAT cs 1to1 2

2018 · Q.12 · 1 mark
Consider the following two tables and four queries in SQL. Book (isbn, bname), Stock (isbn, copies) Query 1: SELECT B.isbn, S.copies FROM Book B INNER JOIN Stock S ON B.isbn = S.isbn; Query 2: SELECT B.isbn, S.copies FROM Book B LEFT OUTER JOIN Stock S ON B.isbn = S.isbn; Query 3: SELECT B.isbn, S.copies FROM Book B RIGHT OUTER JOIN Stock S ON B.isbn = S.isbn; Query 4: SELECT B.isbn, S.copies FROM Book B FULL OUTER JOIN Stock S ON B.isbn = S.isbn; Which one of the queries above is certain to have an output that is a superset of the outputs of the other three queries?
(A) Query 1
(B) Query 2
(C) Query 3
(D) Query 4

2018 · Q.41 · 2 marks
Consider the relations r(A, B) and s(B, C), where s.B is a primary key and r.B is a foreign key referencing s.B. Consider the query Q: r ⋈ (σB<5 (s)) Let LOJ denote the natural left outer-join operation. Assume that r and s contain no null values. Which one of the following queries is NOT equivalent to Q?
(A) σB<5 (r ⋈ s)
(B) σB<5 (r LOJ s)
(C) r LOJ (σB<5 (s))
(D) σB<5 (r) LOJ s CS 14/20 GATE 2018

2017 CS1 · Q.23 · 1 mark [OCR]
Consider a database that has the relation schema EMP (EmpId. EmpName. and DeptName). An instance of the schema EMP and a SQL query on it are given below. EMP EmpId | EmpName | DeptName 1 XYA AA 2 XYB AA 3 XYC AA SELECT AVG(EC.Num) 4 XYD AA FROM EC 5 XYE AB WHERE (DeptName, Num) IN 6 XYF AB (SELECT DeptName, COUNT(EmpId) As - XYG AB EC(DeptName, Num) FROM EMP . 25 aa GROUP BY DeptName) 10 XYJ AC ll XYK AD 12 XYL AD 13 XYM AE The output of executing the SQL query is

2017 CS1 · Q.41 · 2 marks [OCR]
Consider a database that has the relation schemas EMP(EmpId. EmpName. DeptId). and DEPT(DeptName. DeptId). Note that the DeptId can be permitted to be NULL in the relation EMP. Consider the following queries on the database expressed in tuple relational calculus. @ {t|3u ¢ EMP(t{EmpName] = u[EmpName] 4 Vv ¢ DEPT (t[DeptId] = v[DeptId]))} (1) {t| Su ¢ EMP(t{EmpName] = u[EmpName] , Sv ¢ DEPT(t[DeptId] = v[DeptId]))} (Ml) {¢| Ju ¢ EMP(t(EmpName] = u[EmpName] , Sv ¢ DEPT(t(DeptId] = v[DeptId]))} Which of the above queries are safe?
(A) (J) and (ID only
(B) (and (II) only
(C) (ID and (II) only
(D) @. (ID) and (II) =====PAGE 19=====

2017 CS1 · Q.46 · 2 marks [OCR]
Consider a database that has the relation schema CR(StudentName. CourseName). An instance of the schema CR is as given below. CR StudentName CourseName SA CA SA CB SA cc SB CB SB cc sc CA sc CB SE (ge sD CA sD CB sD cc sD cD SE cD SE CA SE CB SF CA SF CB SF CS The following query is made on the database. i 1 1 CourseName (Os GnaName='Sa’ (CR)) T2<CR+H The number of rows in 72 is

2017 CS2 · Q.19 · 1 mark [OCR]
Consider the following tables T1 and T2. Tl T2 P| Q R[S DF || 2 2 [2 3 | 8 8 | 3 7 |3 a. | 2 5 | 8 9 | 7 6 | 9 5 | 7 8 | 5 7 [2 9 | 8 In table T1, P is the primary key and Q is the foreign key referencing R in table T2 with on-delete cascade and on-update cascade. In table T2, R is the primary key and S is the foreign key referencing P in table T1 with on-delete set NULL and on-update cascade. In order to delete record (3,8) from table T1, the number of additional records that need to be deleted from table T1 is

2017 CS2 · Q.46 · 2 marks [OCR]
Consider the following database table named fop_scorer. top_scorer player country goals Klose Germany 16 Ronaldo Brazil 15 GMiiller_ | Germany 14 Fontaine France 13 Pelé Brazil 12 Klinsmann | Germany 11 Kocsis Hungary 11 Batistuta | Argentina 10 Cubillas Peru 10 Lato Poland 10 Lineker England 10 T Miiller | Germany 10 Rahn Germany 10 Consider the following SQL query: SELECT ta.player FROM top scorer AS ta WHERE ta.goals >ALL (SELECT tb.goals FROM top scorer AS tb WHERE tb.country = 'Spain") AND ta.goals >ANY: (SELECT tc.goals FROM top scorer AS tc WHERE tc.country = 'Germany') The number of tuples returned by the above SQL query is

B. New practice questions with solutions

Use these tables for both questions.

R(x, y): (1, a), (2, a), (3, b), (4, c) S(y, z): (a, 10), (a, 20), (b, 30), (d, 40), (NULL, 50)

1. (1 mark · NAT) The number of tuples in the natural join R ⋈ S is _____.

> **5.** y = a gives 2 × 2 = 4 tuples, y = b gives 1, and c has no match. NULL never matches.

2. (2 marks · NAT) The number of rows returned by `SELECT x FROM R WHERE y NOT IN (SELECT y FROM S WHERE z > 15)` is _____.

> **0.** The subquery returns {a, b, d, NULL}. For y = c, the comparison with NULL is UNKNOWN, so `NOT IN` is not true. For a and b the condition is false. No row qualifies.

Normalization & FDs

Candidate keys, closure, BCNF/3NF, dependency preservation. · 17 questions, 27 marks over 15 papers (≈1.8 marks/paper)

A. Past-paper questions (17)

2026 CS1 · Q.30 · 1 mark

Let P, Q, R and S be the attributes of a relation in a relational schema. Let X ⟶ Y indicate functional dependency in the context of a relational database, where X, Y ⊆ {P, Q, R, S}. Which of the following options is/are always true?
(A) If ({P, Q} ⟶ {R} and {P} ⟶ {R}), then {Q} ⟶ {R}
(B) If {P, Q} ⟶ {R}, then ({P} ⟶ {R} or {Q} ⟶ {R})
(C) If ({P} ⟶ {R} and {Q} ⟶ {S}), then {P, Q} ⟶ {R, S}
(D) If {P} ⟶ {R}, then {P, Q} ⟶ {R}

2026 CS1 · Q.31 · 1 mark

statements is/are true? It is always possible to obtain a dependency-preserving 3NF decomposition of a
(A) relation It is always possible to obtain a dependency-preserving 1NF decomposition of a
(B) relation It is not always possible to obtain a dependency-preserving BCNF decomposition
(C) of a relation It is not always possible to obtain a dependency-preserving 2NF decomposition of
(D) a relation

2026 CS1 · Q.65 · 2 marks

Consider a relational database schema with a relation R(A, B, C, D). If {A, B} and {A, C} are the only two candidate keys of the relation R, then the number of superkeys of relation R is _____. (answer in integer)

2026 CS2 · Q.42 · 2 marks

In the context of schema normalization in relational DBMS, consider a set F of functional dependencies. The set of all functional dependencies implied by F is called the closure of F. To compute the closure of F, Armstrong’s Axioms can be applied. Consider X, Y, and Z as sets of attributes over a relational schema. The three rules of Armstrong’s Axioms are described as follows. Reflexivity: If Y ⊆ X , then X → Y Augmentation: If X → Y, then XZ → YZ for any Z Transitivity: If X → Y and Y → Z, then X → Z The additional rule of Union is defined as follows. Union: If X → Y and X → Z, then X → YZ It can be proved that the additional rule of Union is also implied by the three rules of Armstrong’s Axioms. Listed below are four combinations of these three rules. Which one of these combinations is both necessary and sufficient for the proof ?
(A) Reflexivity, Augmentation, and Transitivity
(B) Reflexivity and Augmentation
(C) Transitivity
(D) Augmentation and Transitivity

2025 CS1 · Q.47 · 2 marks
Consider a relational schema team(name, city, owner), with functional dependencies {name → city, name → owner}. The relation team is decomposed into two relations, t1(name, city) and t2(name, owner). Which of the following statement(s) is/are TRUE?
(A) The relation team is NOT in BCNF.
(B) The relations t1 and t2 are in BCNF.
(C) The decomposition constitutes a lossless join.
(D) The relation team is NOT in 3NF.

2025 CS2 · Q.46 · 2 marks
Consider the following relational schema along with all the functional dependencies that hold on them. R1(A, B, C, D, E): { D → E, EA → B, EB → C} R2(A, B, C, D): { A → D, A → B, C → A} Which of the following statement(s) is/are TRUE?
(A) R1 is in 3NF
(B) R2 is in 3NF
(C) R1 is NOT in 3NF
(D) R2 is NOT in 3NF

2024 CS1 · Q.22 · 1 mark
Which of the following statements about a relation R in first normal form (1NF) is/are TRUE ?
(A) R can have a multi-attribute key
(B) R cannot have a foreign key
(C) R cannot have a composite attribute
(D) R cannot have more than one candidate key

2024 CS1 · Q.44 · 2 marks
The symbol → indicates functional dependency in the context of a relational database. Which of the following options is/are TRUE?
(A) (X, Y) → (Z, W) implies X → (Z, W)
(B) (X, Y) → (Z, W) implies (X, Y) → Z
(C) ((X, Y) → Z and W → Y) implies (X, W) → Z
(D) (X → Y and Y → Z) implies X → Z

2024 CS2 · Q.56 · 2 marks
A functional dependency F:X → Y is termed as a useful functional dependency if and only if it satisfies all the following three conditions: • X is not the empty set. • Y is not the empty set. • Intersection of X and Y is the empty set. For a relation R with 4 attributes, the total number of possible useful functional dependencies is _____

2022 · Q.31 · 1 mark
Consider a relation R (A, B , C , D, E) with the following three functional dependencies. AB → C; BC → D; C → E; The number of superkeys in the relation R is _____.

2021 · Q.33 · 2 marks [OCR]
Consider the relation R(P,Q, S,T, X,Y, Z,W) with the following functional depen- dencies. PQ > X; Pa YX; Q3Y; Y> ZW Consider the decomposition of the relation FR into the constituent relations accord- ing to the following two decomposition schemes. Di: = R= ((P.Q,S,T); (P.T,X); (Q.Y); (¥.Z,W)] Dz: R=((P.Q,S); (T,X); (Q.Y); (Y.Z,W)] Which one of the following options is correct?
(A) | D, is a lossless decomposition, but Dy, is a lossy decomposition.
(B) | D, is a lossy decomposition, but D> is a lossless decomposition.
(C) | Both D, and Dy are lossless decompositions.
(D) | Both D; and Dy are lossy decompositions. CS, Set-1 - Copyright © GATE 2021

2020 · Q.36 · 2 marks [OCR]
Consider a relational table R that is in 3NF, but not in BCNF. Which one of the following statements is TRUE?
(A) R has a nontrivial functional dependency X — A, where X is not a superkey and A isa prime attribute.
(B) R has a nontrivial functional dependency — A, where YX is not a superkey and A is anon-prime attribute and X is not a proper subset of any key. © R has a nontrivial functional dependency X — A, where X is not a superkey and A is a non-prime attribute and X is a proper subset of some key. (0) A cell in R holds a set instead of an atomic value.

2019 · Q.32 · 2 marks [OCR]
Let the set of functional dependencies F = {QR > S, R > P, S > Q} hold on a relation schema X = (PQRS). X is not in BCNF. Suppose X is decomposed into two schemas Y and Z, where Y = (PR) and Z = (QRS). Consider the two statements given below. vi Both Y and Z are in BCNF IL. Decomposition of X into Y and Z is dependency preserving and lossless Which of the above statements is/are correct?
(A) Both I and IL
(B) Ionly
(C) ILonly
(D) Neither I nor II

2018 · Q.42 · 2 marks
Consider the following four relational schemas. For each schema, all non-trivial functional dependencies are listed. The underlined attributes are the respective primary keys. Schema I: Registration (rollno, courses) Field ‘courses’ is a set-valued attribute containing the set of courses a student has registered for. Non-trivial functional dependency: rollno courses Schema II: Registration (rollno, courseid, email) Non-trivial functional dependencies: rollno, courseid email email rollno Schema III: Registration (rollno, courseid, marks, grade) Non-trivial functional dependencies: rollno, courseid marks, grade marks grade Schema IV: Registration (rollno, courseid, credit) Non-trivial functional dependencies: rollno, courseid credit courseid credit Which one of the relational schemas above is in 3NF but not in BCNF?
(A) Schema I
(B) Schema II
(C) Schema III
(D) Schema IV

2017 CS1 · Q.16 · 1 mark [OCR]
The following functional dependencies hold true for the relational schema R{V, W,X,Y,Z}: V->-W VW > X Y > VX Yeo Z Which of the following is irreducible equivalent for this set of functional dependencies?
(A) Vow
(B) Vo~W (Cc) Vow
(D) Vow V-xX W-X Vox w-x YoV YovV YoV Y-V Y=3Z YZ Yi X Y :3X% Y-Z Y-Z =====PAGE 10=====

2016 · Q.21 · 1 mark
Which of the following is NOT a superkey in a relational schema with attributes V , W , X, Y , Z and primary key V Y ?
(A) V XY Z
(B) VW XZ
(C) VW XY
(D) VW XY Z

2016 · Q.23 · 1 mark
A database of research articles in a journal uses the following schema. (VOLUME , N UMBER , S TART PAGE , E ND PAGE , T ITLE , Y EAR , P RICE) The primary key is (VOLUME , N UMBER , S TART PAGE , E ND PAGE) and the following functional dependencies exist in the schema. (VOLUME , N UMBER , S TART PAGE , E ND PAGE) → T ITLE (VOLUME , N UMBER) → Y EAR (VOLUME , N UMBER , S TART PAGE , E ND PAGE) → P RICE The database is redesigned to use the following schemas. (VOLUME , N UMBER , S TART PAGE , E ND PAGE , T ITLE , P RICE) (VOLUME , N UMBER , Y EAR) Which is the weakest normal form that the new database satisfies, but the old one does not?
(A) 1NF
(B) 2NF
(C) 3NF
(D) BCNF

B. New practice questions with solutions

1. (1 mark · NAT) R(A, B, C, D, E) has the dependencies A → B, B → C, C → D, D → E, E → A. The number of candidate keys is _____.

5. The dependencies form a cycle, so each single attribute determines all the others.

2. (2 marks · MCQ) R(A, B, C, D) has F = { AB → C, C → D, D → A }. The highest normal form of R is

A. BCNF (B) 3NF but not BCNF (C) 2NF but not 3NF (D) 1NF only

(B). The candidate keys are AB, BC and BD, so every attribute is prime and R is in 3NF. The dependency C → D has a left side that is not a superkey, so R is not in BCNF.

Transactions & concurrency control

Conflict serializability, 2PL, ACID. · 15 questions, 23 marks over 15 papers (≈1.5 marks/paper)

A. Past-paper questions (15)

2026 CS2 · Q.20 · 1 mark

Consider concurrent execution of two transactions T1 and T2 in a DBMS, both of which access a data object A. For these two transactions to not conflict on A, which one of the following statements must be true?
(A) Both T1 and T2 only read A
(B) T1 reads A and T2 writes A
(C) T1 writes A and T2 reads A
(D) Both T1 and T2 write A Consider a file of size 4 million bytes being transferred between two hosts

2025 CS1 · Q.15 · 1 mark

A schedule of three database transactions T1 , T2 , and T3 is shown. Ri
(A) and Wi
(A) denote read and write of data item A by transaction Ti , i = 1,2,3. The transaction T1 aborts at the end. Which other transaction(s) will be required to be rolled back? R1 (X) W1 (Y) R2 (X) R2 (Y) R3 (Y) ABORT(T1)
(A) Only T2
(B) Only T3
(C) Both T2 and T3
(D) Neither T2 nor T3

2025 CS2 · Q.27 · 1 mark

An audit of a banking transactions system has found that on an earlier occasion, two joint holders of account A attempted simultaneous transfers of Rs. 10000 each from account A to account B. Both transactions read the same value, Rs. 11000, as the initial balance in A and were allowed to go through. B was credited Rs. 10000 twice. A was debited only once and ended up with a balance of Rs. 1000. Which of the following properties is/are certain to have been violated by the system?
(A) Atomicity
(B) Consistency
(C) Isolation
(D) Durability

2025 CS2 · Q.53 · 2 marks

Consider the database transactions T1 and T2, and data items X and Y. Which of the schedule(s) is/are conflict serializable? Transaction T1 Transaction T2 R1(X) W2(X) W1(Y) W2(Y) R1(X) COMMIT(T2) W1(X) COMMIT(T1)
(A) R1(X), W2(X), W1(Y), W2(Y), R1(X), W1(X), COMMIT(T2), COMMIT(T1)
(B) W2(X), R1(X), W2(Y), W1(Y), R1(X), COMMIT(T2), W1(X), COMMIT(T1)
(C) R1(X), W1(Y), W2(X), W2(Y), R1(X), W1(X), COMMIT(T1), COMMIT(T2)
(D) W2(X), R1(X), W1(Y), W2(Y), R1(X), COMMIT(T2), W1(X), COMMIT(T1)

2024 CS1 · Q.46 · 2 marks
Consider the following read-write schedule S over three transactions T1 , T2 , and T3 , where the subscripts in the schedule indicate transaction IDs: S: r1 (z); w1 (z); r2 (x); r3 (y); w3 (y); r2 (y); w2 (x); w2 (y); Which of the following transaction schedules is/are conflict equivalent to S ?
(A) T1 T2 T3
(B) T1 T3 T2
(C) T3 T2 T1
(D) T3 T1 T2

2024 CS2 · Q.19 · 1 mark
Once the DBMS informs the user that a transaction has been successfully completed, its effect should persist even if the system crashes before all its changes are reflected on disk. This property is called
(A) durability
(B) atomicity
(C) consistency
(D) isolation

2024 CS2 · Q.27 · 1 mark
Which of the following statements about the Two Phase Locking (2PL) protocol is/are TRUE?
(A) 2PL permits only serializable schedules With 2PL, a transaction always locks the data item being read or written just
(B) before every operation and always releases the lock just after the operation With 2PL, once a lock is released on any data item inside a transaction, no more
(C) locks on any data item can be obtained inside that transaction
(D) A deadlock is possible with 2PL

2022 · Q.39 · 2 marks
Let Ri (z) and Wi (z) denote read and write operations on a data element z by a transaction Ti , respectively. Consider the schedule S with four transactions. S: R4 (x) R2 (x)R3 (x) R1 (y)W1 (y)W2 (x)W3 (y) R4 (y) Which one of the following serial schedules is conflict equivalent to S?
(A) T →T →T →T
(B) T →T →T →T
(C) T →T →T →T
(D) T →T →T →T

2021 · Q.32 · 2 marks [OCR]
Let r;(z) and w;(z) denote read and write operations respectively on a data item z by a transaction T;. Consider the following two schedules. Si: ri(x) ri(y) r2(x) ra(y) we(y) wi(x) Sy: 1ry(x) r2(z) ro(y) we(y) ri(y) wi(z) Which one of the following options is correct?
(A) | S, is conflict serializable, and S, is not conflict serializable.
(B) | S; is not conflict serializable, and S2 is conflict serializable.
(C) | Both S; and S, are conflict serializable.
(D) | Neither S, nor S is conflict serializable.

2020 · Q.37 · 2 marks [OCR]
Consider a schedule of transactions T, and T2: T, | RA RC WD WB | Commit Tz RB | WB RD WC Commit Here, RX stands for “Read(X)” and WX stands for “Write(X)”. Which one of the following schedules is conflict equivalent to the above schedule?
(A) Ty RA|RC}] WD | WB Commit Tz | RB | WB | RD WC Commit (8) T, |RA|RC] WD] WB Commit T2 RB | WB | RD | WC Commit © T, |RA|RC| WD WB Commit Tz RB | WB} RD WC Commit (0) Tr. RA | RC | WD | WB | Commit Tz | RB | WB | RD | WC Commit

2019 · Q.11 · 1 mark [OCR]
Consider the following two statements about database transaction schedules: uf Strict two-phase locking protocol generates conflict serializable schedules that are also recoverable. I. Timestamp-ordering concurrency control protocol with Thomas’ Write Rule can generate view serializable schedules that are not conflict serializable. Which of the above statements is/are TRUE?
(A) Ionly
(B) I only
(C) Both I and I
(D) Neither I nor IT

2017 CS1 · Q.42 · 2 marks [OCR]
In a database system, unique timestamps are assigned to each transaction using Lamport’s logical clock. Let TS(Z,) and TS(TZ,) be the timestamps of transactions J, and J, respectively. Besides. T, holds a lock on the resource R. and J, has requested a conflicting lock on the same resource R. The following algorithm is used to prevent deadlocks in the database system assuming that a killed transaction is restarted with the same timestamp. if TS(Z,) < TS(Z,) then T, is killed else J, waits. Assume any transaction that is not killed terminates eventually. Which of the following is TRUE about the database system that uses the above algorithm to prevent deadlocks?
(A) The database system is both deadlock-free and’staryation-free.
(B) The database system is deadlock-free, but not starvation-free.
(C) The database system is starvation-free. but not deadlock-free.
(D) The database system is neither deadlock-free nor starvation-free.

2017 CS2 · Q.44 · 2 marks [OCR]
Two transactions T, and T, are given as T,:7(X)w,(X)r, (Yi (¥) T2:12(Y)w2(Y)r2(Z)w2(Z) where 7;(V) denotes a read operation by transaction T; on a variable V and w,(V) denotes a write operation by transaction T; on a variable V. The total number of conflict serializable schedules that can be formed by T, and T, is

2016 · Q.22 · 1 mark
Which one of the following is NOT a part of the ACID properties of database transactions?
(A) Atomicity
(B) Consistency
(C) Isolation
(D) Deadlock-freedom

2016 · Q.51 · 2 marks

Consider the following two phase locking protocol. Suppose a transaction T accesses (for read or write operations), a certain set of objects {O1 , . . . , Ok }. This is done in the following manner: Step 1. T acquires exclusive locks to O1 , . . . , Ok in increasing order of their addresses. Step 2. The required operations are performed. Step 3. All locks are released. This protocol will

(A) guarantee serializability and deadlock-freedom

(B) guarantee neither serializability nor deadlock-freedom

(C) guarantee serializability but not deadlock-freedom

(D) guarantee deadlock-freedom but not serializability

B. New practice questions with solutions

1. (1 mark · MCQ) The schedule S = $r_1(A)$ $w_2(A)$ $r_2(B)$ $w_1(B)$ is

A. conflict serializable as T_1 T_2
B. conflict serializable as T_2 T_1
C. not conflict serializable
D. serial

(C). $r_1(A)$ before $w_2(A)$ gives $T_1 \rightarrow T_2$, and $r_2(B)$ before $w_1(B)$ gives $T_2 \rightarrow T_1$. The precedence graph has a cycle.

2. (2 marks · MCQ) The schedule is $r_1(X)$ $w_2(X)$ $r_3(Y)$ $w_1(Y)$ $w_3(Z)$ $r_2(Z)$. It is conflict equivalent to the serial schedule

A. T_1 T_2 T_3 (B) T_3 T_1 T_2 (C) T_3 T_2 T_1 (D) T_2 T_1 T_3

(B). The conflict edges are $T_1 \rightarrow T_2$ (on X), $T_3 \rightarrow T_1$ (on Y) and $T_3 \rightarrow T_2$ (on Z). The only topological order is T_3, T_1, T_2.

Indexing, B/B+ trees, file organization

B/B+ trees, file organization, blocking factor. · 9 questions, 14 marks over 15 papers (≈0.9 marks/paper)

A. Past-paper questions (9)

2026 CS2 · Q.46 · 2 marks

search-key value in the indexed file. Otherwise it is called a sparse index. Consider the following two statements. S1: A hash index must be a dense index S2: A B + tree index can be a sparse index Which one of the following options is correct?
(A) Both S1 and S2 are true
(B) Both S1 and S2 are false
(C) S1 is true and S2 is false
(D) S1 is false and S2 is true

2025 CS1 · Q.21 · 1 mark

Consider the following B + tree with 5 nodes, in which a node can store at most 3 key values. The value 23 is now inserted in the B + tree. Which of the following options(s) is/are CORRECT?
(A) None of the nodes will split.
(B) At least one node will split and redistribute.
(C) The total number of nodes will remain same.
(D) The height of the tree will increase.

2025 CS2 · Q.57 · 2 marks

In a B+- tree where each node can hold at most four key values, a root to leaf path consists of the following nodes: A = (49, 77, 83, -), B = (7, 19, 33, 44), C = (20*, 22*, 25*, 26*) The *-marked keys signify that these are data entries in a leaf. Assume that a pointer between keys k1 and k2 points to a subtree containing keys in [k1 , k2), and that when a leaf is created, the smallest key in it is copied up into its parent. A record with key value 23 is inserted into the B+- tree. The smallest key value in the parent of the leaf that contains 25* is _____. (Answer in integer)

2024 CS1 · Q.21 · 1 mark

In a B+ tree, the requirement of at least half-full (50%) node occupancy is relaxed for which one of the following cases?
(A) Only the root node
(B) All leaf nodes
(C) All internal nodes
(D) Only the leftmost leaf node

2024 CS2 · Q.26 · 1 mark

Which of the following file organizations is/are I/O efficient for the scan operation in DBMS?
(A) Sorted
(B) Heap
(C) Unclustered tree index
(D) Unclustered hash index

2023 · Q.62 · 2 marks
Consider a database of fixed-length records, stored as an ordered file. The database has 25,000 records, with each record being 100 bytes, of which the primary key occupies 15 bytes. The data file is block-aligned in that each data record is fully contained within a block. The database is indexed by a primary index file, which is also stored as a block-aligned ordered file. The figure below depicts this indexing scheme. Data File Primary Key Other Fields (15 Bytes) (85 Bytes) Index File Block Anchor Block Primary Key Pointer Suppose the block size of the file system is 1024 bytes, and a pointer to a block occupies 5 bytes. The system uses binary search on the index file to search for a record with a given key. You may assume that a binary search on an index file of b blocks takes log2 b block accesses in the worst case. Given a key, the number of block accesses required to identify the block in the data file that may contain a record with the key, in the worst case, is . CS

2020 · Q.54 · 2 marks [OCR]
Consider a database implemented using B+ tree for file indexing and installed on a disk drive with block size of 4 KB. The size of search key is 12 bytes and the size of tree/disk pointer is 8 bytes. Assume that the database has one million records. Also assume that no node of the B+ tree and no records are present initially in main memory. Consider that each record fits into one disk block. The minimum number of disk accesses required to retrieve any record in the database is

2019 · Q.14 · 1 mark [OCR]
Which one of the following statements is NOT correct about the B+ tree data structure used for creating an index of a relational database table?
(A) B+ Tree is a height-balanced tree
(B) Non-leaf nodes have pointers to data records
(C) Key values in each node are kept in sorted order
(D) Each leaf node has a pointer to the next leaf node 243k pi0+i2zk ,

2017 CS2 · Q.49 · 2 marks [OCR]
In a B+ tree. if the search-key value is 8 bytes long. the block size is 512 bytes and the block pointer size is 2 bytes, then the maximum order of the B+ tree is

B. New practice questions with solutions

1. (1 mark · NAT) A B+ tree has 1024-byte nodes, 8-byte keys and 6-byte block pointers. The order p of an internal node, the largest number of pointers per node satisfying $p \cdot 6 + (p - 1) \cdot 8 \le 1024$, is _____.

73. $14p \le 1032$ gives $p \le 73.7$, so $p = 73$.

2. (2 marks · NAT) An ordered file has 100,000 records of 100 bytes in 4096-byte blocks, with no spanning. A sparse primary index has 16-byte entries. The number of block accesses to fetch one record by binary search on the index file is _____.

5. The blocking factor is 40, giving 2500 data blocks. The sparse index has 2500 entries at 256 per block, so 10 index blocks. Binary search takes $\lceil \log_2 10 \rceil = 4$ accesses, plus 1 data block.

ER model

Cardinality, weak entities, ER model. · 6 questions, 6 marks over 15 papers (≈0.4 marks/paper)

A. Past-paper questions (6)

2026 CS2 · Q.15 · 1 mark
In the context of DBMS, consider the two sets T and S given below. T S I: Logical schema L: Views II: Physical schema M: File organization and indexes III: External schema N: Relations Which one of the following is the correct match from T to S ?
(A) I – L, II – M, III – N
(B) I – M, II – L, III – N
(C) I – N, II – M, III – L
(D) I – N, II – L, III – M

2024 CS1 · Q.20 · 1 mark
Let S be the specification: "Instructors teach courses. Students register for courses. Courses are allocated classrooms. Instructors guide students." Which one of the following ER diagrams CORRECTLY represents S?
(A) (i)
(B) (ii)
(C) (iii)
(D) (iv)

2024 CS2 · Q.20 · 1 mark
In the context of owner and weak entity sets in the ER (Entity-Relationship) data model, which one of the following statements is TRUE?
(A) The weak entity set MUST have total participation in the identifying relationship
(B) The owner entity set MUST have total participation in the identifying relationship Both weak and owner entity sets MUST have total participation in the identifying
(C) relationship Neither weak entity set nor owner entity set MUST have total participation in the
(D) identifying relationship

2020 · Q.14 · 1 mark [OCR]
Which one of the following is used to represent the supporting many-one relationships of a weak entity set in an entity-relationship diagram?
(A) Diamonds with double/bold border (8) Rectangles with double/bold border (©) Ovals with double/bold border (0) Ovals that contain underlined identifiers

2018 · Q.11 · 1 mark
In an Entity-Relationship (ER) model, suppose R is a many-to-one relationship from entity set E1 to entity set E2. Assume that E1 and E2 participate totally in R and that the cardinality of E1 is greater than the cardinality of E2. Which one of the following is true about R?
(A) Every entity in E1 is associated with exactly one entity in E2.
(B) Some entity in E1 is associated with more than one entity in E2.
(C) Every entity in E2 is associated with exactly one entity in E1.
(D) Every entity in E2 is associated with at most one entity in E1. CS 3/20 GATE 2018

> **2017 CS2 · Q.17 · 1 mark** [OCR]
> An ER model of a database consists of entity types A and B. These are connected by a relationship R which does not have its own attribute. Under which one of the following conditions. can the relational table for R be merged with that of A?
> (A) Relationship R is one-to-many and the participation of A in R is total.
> (B) Relationship R is one-to-many and the participation of A in R is partial.
> (C) Relationship R is many-to-one and the participation of A in R is total.
> (D) Relationship R is many-to-one and the participation of A in R is partial.

B. New practice questions with solutions

1. (1 mark · NAT) Entities E1 (key a) and E2 (key b) are in an M:N relationship with an attribute c. The minimum number of relations in the relational schema is _____.

> **3.** One table per entity, plus one table for the M:N relationship.

2. (2 marks · NAT) An ER diagram has entities A, B, C. R is M:N between A and B. S is 1:N from B to C. T is 1:1 between A and C. The minimum number of relations is _____.

> **4.** Tables for A, B and C, plus one for R. S merges into C as a foreign key to B, and T merges into A or C as a foreign key.